SAECOIN (SAC)

Technical Whitepaper — Draft v1.0

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Consensus Model: Proof of Work (Proposed)
Transaction Model: UTXO (Proposed)
Blockchain Type: Independent Layer-1 Blockchain

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1. Abstract

SAECOIN is a proposed independent Layer-1 blockchain project designed around Proof of Work (PoW) consensus and the Unspent Transaction Output (UTXO) transaction model.

The project aims to develop a cryptocurrency network that supports independent node operation, transparent transaction verification, and CPU/GPU-based mining.

Unlike tokens deployed on existing blockchain networks, SAECOIN is intended to operate through its own blockchain infrastructure, including its own consensus rules, peer-to-peer network, transaction system, block production mechanism, and wallet ecosystem.

The project prioritizes:

- Independent blockchain infrastructure.
- Transparent and verifiable transactions.
- Open-source development.
- Accessible CPU and GPU mining.
- Secure wallet architecture.
- Reproducible network rules.
- Long-term maintainability.

All technical parameters in this document are proposed design choices and remain subject to protocol review, mathematical verification, security testing, and community evaluation.

2. Vision and Mission

2.1 Vision

To develop an independent, transparent, and accessible Proof of Work cryptocurrency network that allows individuals to participate in transaction verification and block production.

2.2 Mission

SAECOIN aims to:

1. Build an independent Layer-1 blockchain.
2. Develop a reliable Proof of Work consensus mechanism.
3. Support CPU and GPU mining.
4. Provide secure wallet applications.
5. Enable independent full-node operation.
6. Publish open-source software and protocol documentation.
7. Establish a carefully tested network before any mainnet launch.

3. Problem Statement

Many cryptocurrency projects depend on existing blockchain networks or operate through centrally controlled infrastructure.

SAECOIN explores an alternative approach: creating a separate blockchain with its own consensus rules and network infrastructure.

The project focuses on several design challenges:

- Making blockchain participation accessible.
- Maintaining reliable transaction verification.
- Preventing unauthorized creation of coins.
- Reducing dependence on centralized infrastructure.
- Protecting wallet keys and transaction integrity.
- Maintaining predictable and verifiable monetary rules.

SAECOIN does not claim that these challenges have already been solved. They are engineering objectives that must be demonstrated through implementation and testing.

4. Core Design Principles

SAECOIN development will follow these principles:

4.1 Independence

SAECOIN is intended to have its own blockchain rather than being issued as a token on another network.

4.2 Transparency

Protocol rules, source code, transaction validation, and network parameters should be publicly inspectable.

4.3 Verifiability

Every full node should independently validate blocks and transactions according to the same consensus rules.

4.4 Mining Accessibility

CPU and GPU mining are design objectives. The final mining algorithm must be evaluated for hardware accessibility and resistance to specialized mining dominance.

4.5 Security First

Consensus implementation, cryptographic operations, wallet storage, and network communication must undergo extensive testing and independent review.

4.6 Open Development

The project intends to use public source-code repositories and documented development processes.

5. Blockchain Architecture

SAECOIN is proposed as an independent Layer-1 blockchain.

Its architecture consists of the following major components:

5.1 Blockchain Core

The blockchain core will handle:

- Block validation.
- Transaction validation.
- Consensus verification.
- Chain selection.
- Block reward calculation.
- Difficulty adjustment.
- UTXO state management.

5.2 Full Nodes

Full nodes will:

- Maintain blockchain data.
- Validate received blocks.
- Verify transactions.
- Communicate with other nodes.
- Reject invalid consensus data.
- Provide network services where enabled.

5.3 Miners

Miners will:

- Collect valid transactions.
- Construct candidate blocks.
- Perform Proof of Work.
- Submit valid blocks to the network.

5.4 Wallets

Wallet applications are planned for:

- Android.
- Windows desktop.

Wallet functionality is expected to include:

- Wallet creation.
- Key generation.
- Backup and recovery.
- Address management.
- Transaction creation.
- Transaction history.
- Balance calculation.
- Transaction status monitoring.

5.5 Block Explorer

A web-based explorer is planned to provide publicly accessible information such as:

- Blocks.
- Transactions.
- Addresses.
- Network statistics.

- Mining information.

An explorer will display blockchain data but will not replace independent node verification.

6. Consensus Mechanism

6.1 Proof of Work

SAECOIN proposes using Proof of Work as its consensus mechanism.

Under this model, miners compete to find a valid block hash that satisfies the network's current difficulty target.

A valid block must satisfy all applicable consensus rules, including:

- Valid Proof of Work.
- Correct previous-block reference.
- Valid transaction structure.
- Correct transaction ordering and accounting rules.
- Valid block reward.
- Valid timestamp and difficulty rules.

Full nodes independently verify these conditions before accepting a block.

6.2 CPU and GPU Mining

SAECOIN intends to support CPU and GPU mining.

The final Proof of Work algorithm has not yet been selected.

Algorithm selection must consider:

- Security.
- Mining performance.
- CPU compatibility.
- GPU compatibility.
- Memory requirements.
- Energy efficiency.
- Specialized hardware advantages.
- Long-term maintenance.

ASIC resistance is a design objective, not a guaranteed property. No algorithm can be declared ASIC-resistant without careful analysis and ongoing evaluation.

6.3 Difficulty Adjustment

The network will require a difficulty adjustment mechanism to maintain a reasonably stable block production interval.

The exact difficulty formula, adjustment window, limits, and protection against manipulation remain undecided.

These rules must be finalized before consensus implementation.

7. Transaction Model

SAECOIN proposes using the UTXO model.

UTXO stands for Unspent Transaction Output.

Instead of maintaining a conventional account balance for every address, the blockchain tracks transaction outputs that have not yet been spent.

A transaction consumes existing UTXOs and creates new outputs.

The validation system must ensure:

- Inputs reference existing unspent outputs.
- Inputs are authorized by valid signatures.
- No output is spent more than once.
- Transaction values are conserved.
- Fees are calculated correctly.
- Invalid transactions are rejected.

This model is intended to provide transparent transaction accounting and independent verification.

8. Proposed Monetary Policy

The following parameters represent the current proposed economic design.

They are not yet final consensus rules.

Parameter	Proposed Value
Coin Name	SAECOIN
Ticker	SAC (Provisional)
Maximum Supply Target	21,000,000 SAC
Initial Block Subsidy	50 SAC

Target Block Time	10 Minutes
Proposed Halving Interval	210,000 Blocks
Premine	0 SAC
Decimal Precision	8
Smallest Unit	0.00000001 SAC

8.1 Maximum Supply

The current design target is a maximum supply of 21 million SAC.

The final issuance schedule must be mathematically verified before implementation.

In particular, the proposed initial subsidy and halving interval do not automatically prove that total issuance will equal exactly 21 million SAC.

The protocol team must define:

- Subsidy reduction rules.
- Integer rounding behavior.
- Final subsidy conditions.
- Total lifetime issuance.
- Handling of fees and unclaimed rewards.

The final supply calculation must be independently reviewed and documented.

8.2 Block Reward

The proposed initial block subsidy is 50 SAC.

Miners who produce valid blocks may receive the permitted block subsidy and applicable transaction fees.

The network must enforce the reward limit through consensus validation.

8.3 Halving

A reduction in block subsidy is proposed every 210,000 blocks.

The exact schedule and resulting total issuance remain subject to mathematical verification.

8.4 Premine

The current proposal is zero premine.

No coins are intended to be created or distributed outside the publicly defined consensus issuance rules.

The genesis block and initial network state must be reproducible and independently verifiable.

9. Network Architecture

SAECOIN will require a peer-to-peer network.

Nodes will communicate to discover peers and exchange blockchain data.

The network design must address:

- Peer discovery.
- Block propagation.
- Transaction propagation.
- Message validation.
- Connection limits.
- Malicious peer behavior.
- Network abuse prevention.
- Synchronization.
- Chain reorganization.

The network protocol and message formats remain subject to specification and implementation review.

10. Wallet Architecture

Wallet security is a critical component of SAECOIN.

The wallet system is expected to use secure cryptographic key generation and signing.

Planned wallet applications:

Android Wallet

Technology direction:

- Kotlin.
- Jetpack Compose.

Windows Desktop Wallet

Technology direction:

- Tauri.
- Rust.
- TypeScript.

Wallet security requirements include:

- Secure private-key generation.
- Encrypted local key storage.
- Backup and recovery.
- Protection against unauthorized access.
- Safe transaction signing.
- Clear transaction confirmation screens.

The final key algorithm, seed format, derivation standard, address encoding, and encryption parameters must be selected and reviewed before implementation.

11. Technology Stack

The proposed technology stack is:

Component	Proposed Technology
Blockchain Core	Rust
Node Software	Rust
Consensus Engine	Rust
Transaction Engine	Rust
P2P Networking	Rust
Mining Core	Rust
Wallet Core	Rust
Android Wallet	Kotlin + Jetpack Compose
Windows Wallet	Tauri + Rust + TypeScript
Explorer Frontend	React + TypeScript
Explorer Backend	To Be Finalized
Database	RocksDB (Provisional)
Source Control	Git + GitHub

Technology choices may change after benchmarking and architecture review.

12. Security Model

SAECOIN's security will depend on correct implementation, network participation, cryptographic safety, and independent validation.

Potential security risks include:

- Majority hash-power attacks.
- Double-spending attempts.
- Selfish mining.
- Network partitioning.
- Eclipse attacks.
- Invalid block construction.
- Transaction malleability.
- Wallet key theft.
- Software supply-chain compromise.
- Consensus implementation bugs.

The project will require:

1. Unit testing.
2. Integration testing.
3. Consensus testing.
4. Fuzz testing.
5. Adversarial testing.
6. Wallet recovery testing.
7. Network stress testing.
8. Independent security review.

Proof of Work does not guarantee absolute security. Security depends on the network's actual hash-power distribution and implementation quality.

13. Development Roadmap

The proposed development stages are:

Phase 1 — Documentation

- Complete project requirements.
- Define blockchain architecture.
- Document consensus requirements.
- Document security requirements.
- Define data structures.
- Identify unresolved protocol decisions.

Phase 2 — Protocol Finalization

- Select Proof of Work algorithm.
- Finalize monetary issuance mathematics.
- Define difficulty adjustment.
- Define block structure.
- Define transaction rules.
- Define cryptographic standards.
- Define genesis block parameters.

Phase 3 — Blockchain Core

- Implement blockchain data structures.
- Implement transaction validation.
- Implement UTXO management.
- Implement consensus validation.
- Implement block processing.

Phase 4 — Node and P2P Network

- Implement full-node operation.
- Implement peer discovery.
- Implement block propagation.
- Implement synchronization.
- Test network reliability.

Phase 5 — Mining

- Implement CPU mining.
- Develop GPU mining support.
- Benchmark mining performance.
- Evaluate hardware accessibility.
- Test mining stability.

Phase 6 — Wallet Development

- Develop Android wallet.
- Develop Windows wallet.
- Implement secure key management.
- Implement backup and recovery.
- Integrate transaction functionality.

Phase 7 — Explorer

- Develop explorer backend.
- Develop explorer frontend.
- Display blocks and transactions.
- Add network statistics.

Phase 8 — Testnet

- Launch private testnet.
- Perform public testnet testing.
- Test mining distribution.
- Test network attacks.
- Test wallet recovery.
- Fix identified issues.

Phase 9 — Independent Review

- Conduct protocol review.
- Conduct mathematical review.
- Conduct security review.
- Resolve critical issues.
- Publish review findings.

Phase 10 — Mainnet Readiness

Mainnet launch may be considered only after:

- Consensus rules are finalized.
- Genesis block is reproducible.
- Independent reviews are completed.
- Critical bugs are resolved.
- Testnet results are acceptable.
- Release binaries are reproducible.
- Operational procedures are documented.

No mainnet launch date is promised in this draft.

14. Governance and Protocol Upgrades

The final governance model has not yet been selected.

Future decisions must establish:

- How protocol changes are proposed.
- How changes are reviewed.
- How releases are approved.

- How compatibility is maintained.
- How emergency security updates are handled.
- How node operators are informed.

Consensus-changing upgrades must be carefully coordinated and independently reviewed.

15. Risks and Limitations

SAECOIN is an experimental blockchain project proposal.

Potential risks include:

- Technical implementation failure.
- Consensus vulnerabilities.
- Insufficient mining participation.
- Concentration of hash power.
- Network instability.
- Wallet security failures.
- Limited adoption.
- Regulatory uncertainty.
- Infrastructure and maintenance costs.

CPU/GPU mining support does not guarantee fair mining distribution or profitability.

A fixed supply target does not guarantee market value.

No claim is made that SAECOIN will achieve a particular market price, exchange listing, adoption level, or financial return.

16. Legal and Regulatory Considerations

Cryptocurrency regulations differ between countries and may change over time.

Before operating, distributing, marketing, or listing SAECOIN, the project must obtain appropriate legal and regulatory advice for its intended jurisdictions.

This document is not legal, financial, or investment advice.

17. Current Project Status

At the time of this document's preparation:

- SAECOIN is a proposed blockchain project.
- The consensus algorithm has not been finalized.

- The monetary issuance schedule has not been mathematically finalized.
- The cryptographic standards have not been finalized.
- The genesis block has not been generated.
- The blockchain software is not represented as implemented.
- No operational mainnet is being claimed.
- No circulating SAC supply is being claimed.

All proposed features must be implemented, tested, and independently reviewed before they can be considered operational.

18. Conclusion

SAECOIN aims to become an independent Proof of Work blockchain with a UTXO transaction model and support for CPU and GPU mining.

The project emphasizes transparency, independent verification, secure wallet design, and careful protocol development.

Its success will depend on sound engineering, mathematical correctness, security testing, open development, and meaningful network participation.

The immediate priority is not launching a cryptocurrency quickly. It is designing and validating a reliable protocol before implementation and public deployment.

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